Homework

Homework 5 — Splitting an area — 55 min, after S79.

Consider the function f defined on $[0, \frac{\pi}{2}]$ by $f(x) = \sin x \cos x$. Its curve is drawn in an orthonormal frame.

1. Calculate $f'(x)$ and determine the maximum of f on this interval.
2. Solve on $[0, \frac{\pi}{2}]$ the inequality $f(x) \geq \frac{1}{4}$.
3. Calculate $\displaystyle\int_0^{\frac{\pi}{2}} f(x)\ \mathrm{d}x$, then the mean value of f on $[0, \frac{\pi}{2}]$.
4. For $a \in [0, \frac{\pi}{2}]$, let $A(a) = \displaystyle\int_0^a f(x)\ \mathrm{d}x$. Determine a so that the region under the curve between 0 and a has an area equal to half the total area under the curve between 0 and $\frac{\pi}{2}$.